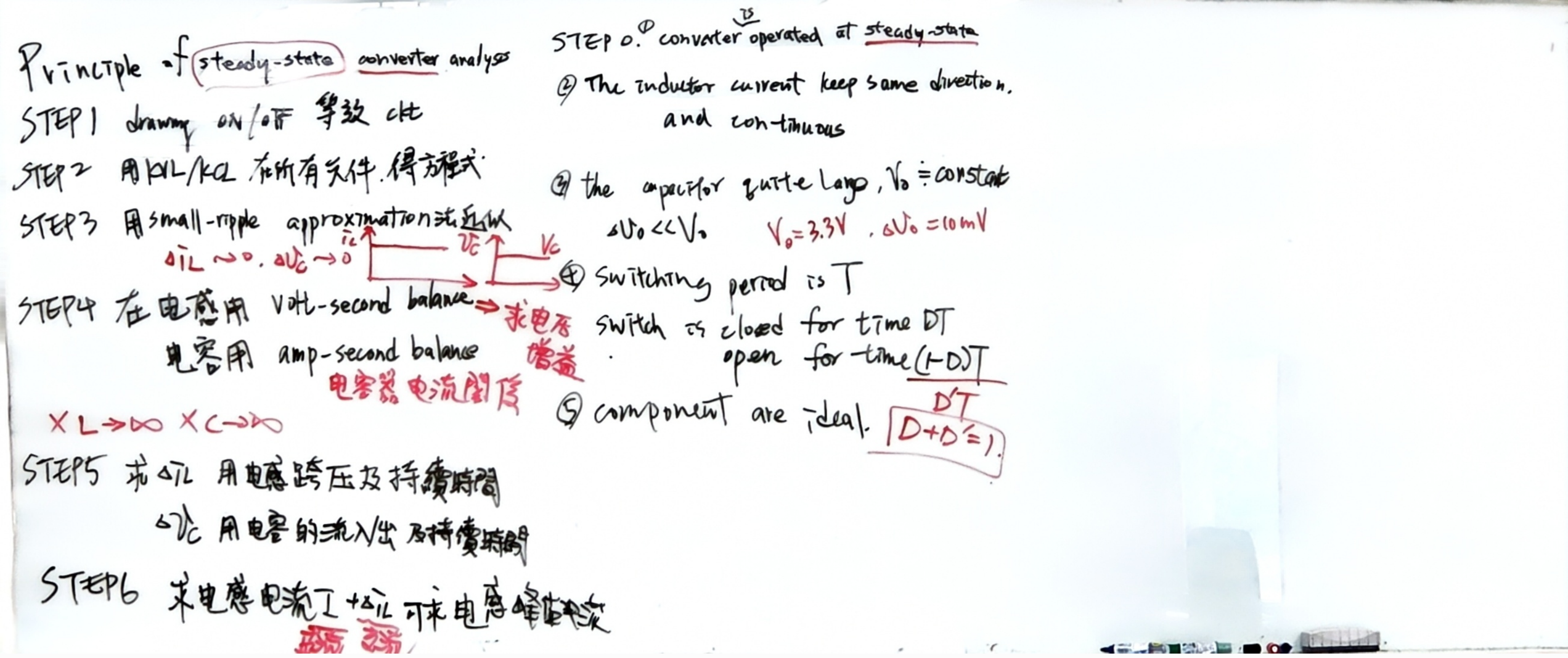


Principle of steady-state converter analysis

STEP1 drawing ON/OFF 等效 ckt

STEP2 用KVL/KCL 在所有元件. 得方程式

STEP3 用small-ripple approximation法近似

$\Delta i_L \to 0$. $\Delta v_C \to 0$

STEP4 在电感用 Volt-second balance ⇒ 求电压增益

电容用 amp-second balance 电容器电流關係

$X_L \to \infty$ $X_C \to \infty$

STEP5 求 Δi_L 用电感跨压及持續時間

Δv_C 用电容的流入/出 及持續時間

STEP6 求电感电流 $I + \Delta i_L$ 可求电感峰值电流

[illegible]

STEP 0. ① converter is operated at steady-state

② The inductor current keep same direction, and continuous

③ the capacitor quite large, $V_0 \fallingdotseq$ constant

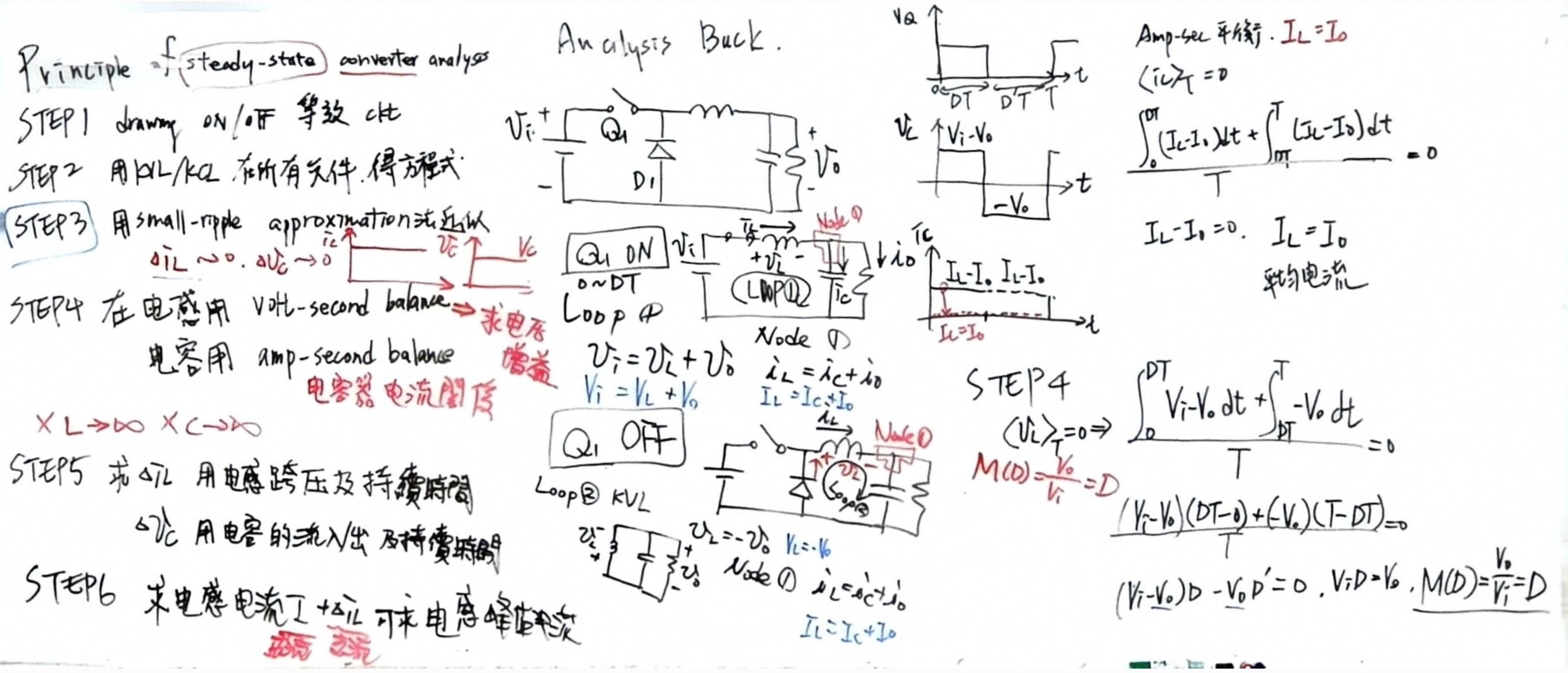
$\Delta V_0 << V_0$ $V_0 = 3.3V$, $\Delta V_0 = 10mV$

④ Switching period is T

switch is closed for time DT

open for time $(1-D)T$ = $D'T$

⑤ component are ideal. $\boxed{D+D'=1}$



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Analysis Buck.

V_i
Q_1
D_1
V_0
V_Q
DT
D'T
T
t
V_L
$V_i - V_0$
$-V_0$
i_C
$I_L - I_0$
$I_L - I_0$
$I_L = I_0$
Node ①
LOOP①
i_L
i_C
i_0
Loop②

$\boxed{Q_1 \text{ ON}}$ 0~DT

Loop ①

$v_i = v_L + v_0$

$V_i = V_L + V_0$

Node ①

$i_L = i_C + i_0$

$I_L = I_C + I_0$

$\boxed{Q_1 \text{ OFF}}$

Loop② KVL

$v_L = -v_0$ $V_L = -V_0$

Node ① $i_L = i_C + i_0$

$I_L = I_C + I_0$

Amp-sec 平衡. $I_L = I_0$

$\langle i_C \rangle_T = 0$

$$\frac{\int_0^{DT} (I_L - I_0)\,dt + \int_{DT}^{T} (I_L - I_0)\,dt}{T} = 0$$

$I_L - I_0 = 0$. $I_L = I_0$

平均电流

STEP4

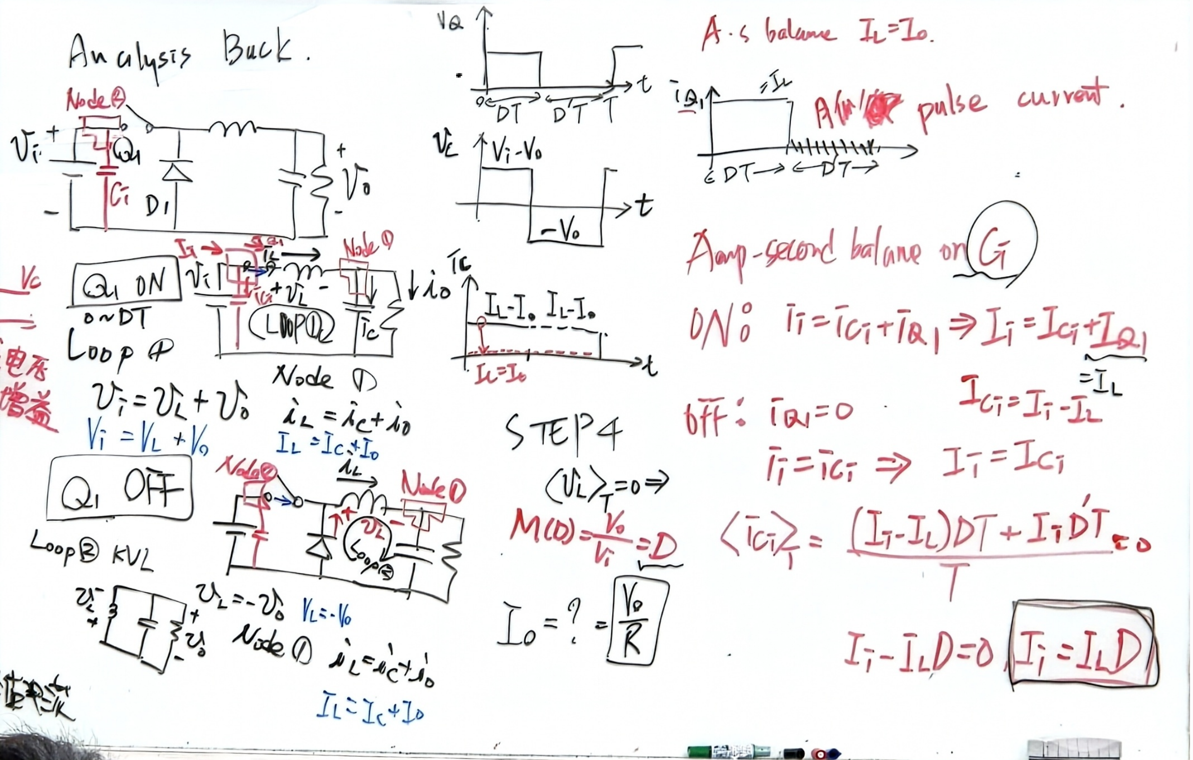
$\langle v_L \rangle_T = 0 \Rightarrow$

$$\frac{\int_0^{DT} V_i - V_0\,dt + \int_{DT}^{T} -V_0\,dt}{T} = 0$$

$M(D) = \frac{V_0}{V_i} = D$

$$\frac{(V_i - V_0)(DT - 0) + (-V_0)(T - DT)}{T} = 0$$

$(V_i - V_0)D - V_0 D' = 0$, $V_i D = V_0$, $M(D) = \frac{V_0}{V_i} = D$



Analysis Buck.

$\boxed{Q_1 \text{ ON}}$ 0~DT

Loop ①

$v_i = v_L + v_0$

$V_i = V_L + V_0$

Node ①

$i_L = i_C + i_0$

$I_L = I_C + I_0$

$\boxed{Q_1 \text{ OFF}}$

Loop② KVL

$v_L = -v_0$ $V_L = -V_0$

Node ① $i_L = i_C + i_0$

$I_L = I_C + I_0$

STEP4

$\langle v_L \rangle_T = 0 \Rightarrow$

$M(D) = \frac{V_0}{V_i} = D$

Node②
C_i
I_i
i_{Q_1}
i_{C_i}
$= I_L$
DT
D'T

$I_0 = ? = \boxed{\frac{V_0}{R}}$

A·s balance $I_L = I_0$.

A[illegible] pulse current.

Amp-second balance on C_i

ON: $i_i = i_{C_i} + i_{Q_1} \Rightarrow I_i = I_{C_i} + \underbrace{I_{Q_1}}_{= I_L}$

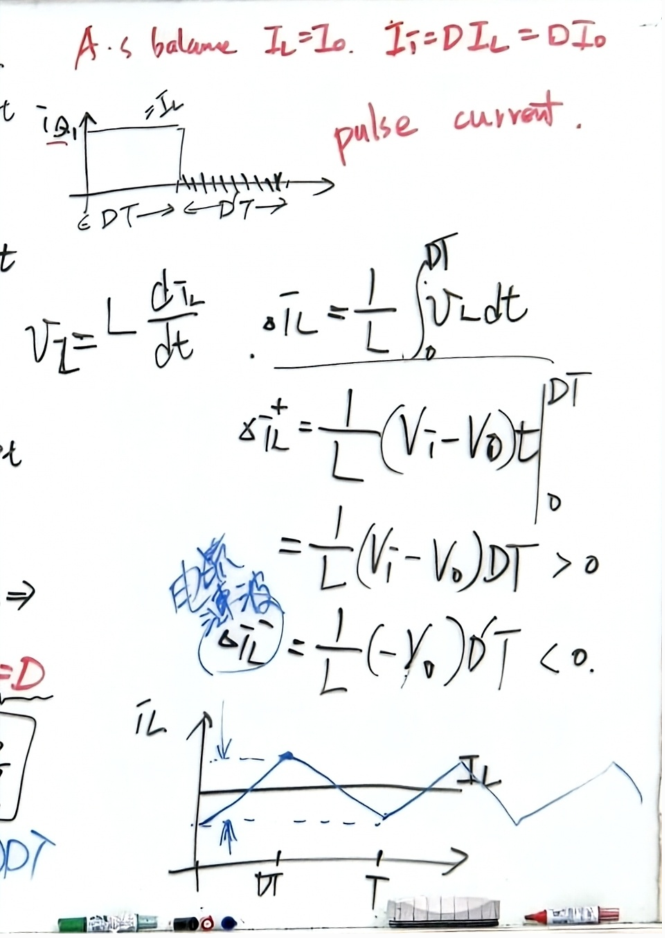
$I_{C_i} = I_i - I_L$

OFF: $i_{Q_1} = 0$

$i_i = i_{C_i} \Rightarrow I_i = I_{C_i}$

$$\langle i_{C_i} \rangle_T = \frac{(I_i - I_L)DT + I_i D'T}{T} = 0$$

$I_i - I_L D = 0$, $\boxed{I_i = I_L D}$


i_{Q1}
$=I_L$
DT
D'T
i_L
I_L
DT
T

$A\cdot s$ balance $I_L = I_o$. $I_i = DI_L = DI_o$

pulse current.

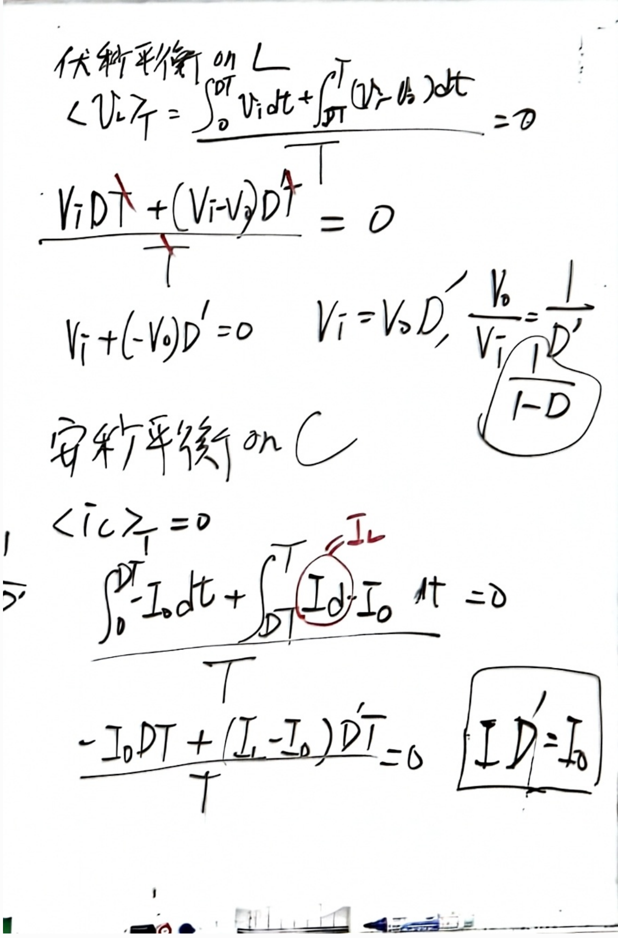
$v_L = L\frac{di_L}{dt}$ $\Delta i_L = \frac{1}{L}\int_0^{DT} v_L dt$

$\Delta i_L^+ = \frac{1}{L}(V_i - V_o)t\Big|_0^{DT}$

$= \frac{1}{L}(V_i - V_o)DT > 0$

电感涟波

$\Delta i_L^- = \frac{1}{L}(-V_o)D'T < 0.$



伏秒平衡 on L

$\langle v_L\rangle_T = \frac{\int_0^{DT} v_i dt + \int_{DT}^{T}(v_i - v_o)dt}{T} = 0$

$\frac{V_iD\cancel{T} + (V_i - V_o)D'\cancel{T}}{\cancel{T}} = 0$

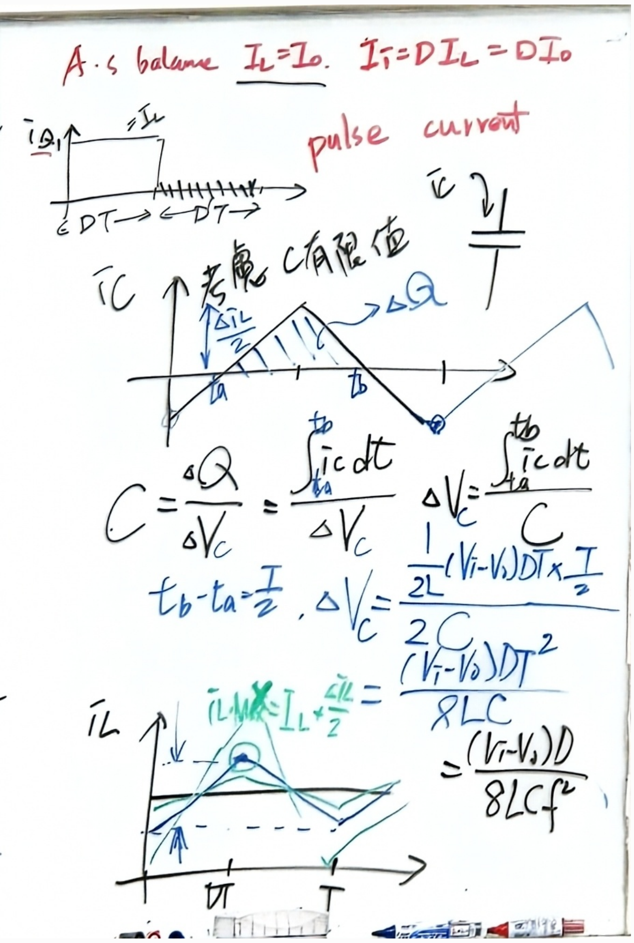
$V_i + (-V_o)D' = 0$ $V_i = V_oD'$, $\frac{V_o}{V_i} = \frac{1}{D'}$ $\frac{1}{1-D}$

安秒平衡 on C

$\langle i_C\rangle_T = 0$

$\frac{\int_0^{DT} -I_o dt + \int_{DT}^{T} (\overset{=I_L}{I_d} - I_o) dt}{T} = 0$

$\frac{-I_oDT + (I_L - I_o)D'T}{T} = 0$ $\boxed{ID' = I_o}$


i_{Q1}
$=I_L$
DT
D'T
i_C
$\frac{\Delta i_L}{2}$
ΔQ
t_a
t_b
i_L
DT
T

$A\cdot s$ balance $I_L = I_o$. $I_i = DI_L = DI_o$

pulse current

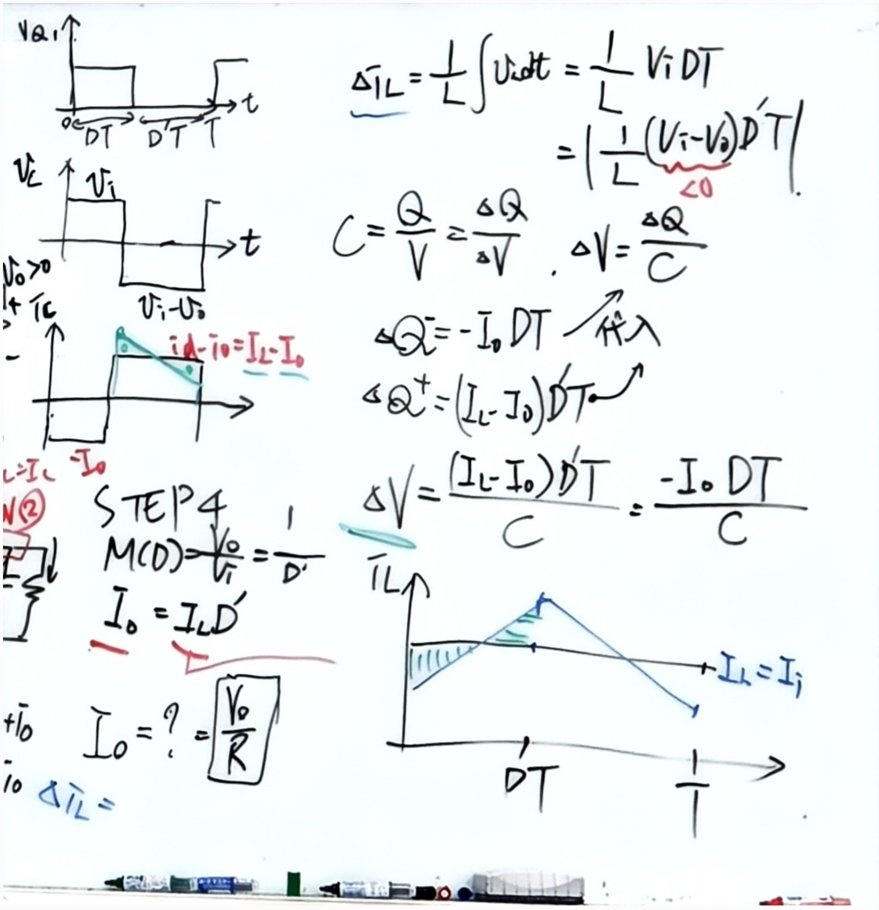
考慮C有限值

$C = \frac{\Delta Q}{\Delta V_C} = \frac{\int_{t_a}^{t_b} i_C dt}{\Delta V_C}$ $\Delta V_C = \frac{\int_{t_a}^{t_b} i_C dt}{C}$

$t_b - t_a = \frac{T}{2}$, $\Delta V_C = \frac{\frac{1}{2L}(V_i - V_o)DT \times \frac{T}{2}}{2C} = \frac{(V_i - V_o)DT^2}{8LC}$

$i_{L\cdot\cancel{\mathrm{Max}}} = I_L + \frac{\Delta i_L}{2}$

$= \frac{(V_i - V_o)D}{8LCf^2}$


v_{Q1}
t
0
DT
D'T
T
v_L
v_i
t
$v_i - v_o$
$V_o > 0$
i_C
$i_d - i_o = I_L - I_o$
$-I_o$
i_L
$I_L = I_i$
DT
T

$\Delta i_L = \frac{1}{L}\int v_i dt = \frac{1}{L}V_iDT$

$= \left|\frac{1}{L}\underbrace{(V_i - V_o)}_{<0}D'T\right|$

$C = \frac{Q}{V} = \frac{\Delta Q}{\Delta V}$, $\Delta V = \frac{\Delta Q}{C}$

$\Delta Q^- = -I_oDT$ 代入

$\Delta Q^+ = (I_L - I_o)D'T$

$\Delta V = \frac{(I_L - I_o)D'T}{C} = \frac{-I_oDT}{C}$

STEP4

$M(D) = \frac{V_o}{V_i} = \frac{1}{D'}$

$I_o = I_LD'$

$I_o = ? = \boxed{\frac{V_o}{R}}$

$\Delta i_L =$